Intrusion Detection and Prevention

Segurança em Redes de Comunicações

Mestrado em Cibersegurança

Mestrado em Engenharia de Computadores e Telemática

DETI-UA

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Intrusion Detection and Prevention

- Intrusion Detection Systems (IDS)
  - Monitoring and identifying unauthorized system access or manipulation.
  - Analyzes information from multiple sources (computers, servers, services, and network traffic).
  - Identifies:
    - Intrusions, attacker outside of the organization;
    - Misuse, wrong behavior from a licit user/service.
  - Does not block/prevent intrusion.
  - Signals an alarm for:
    - Human analysis and intervention;
    - Automatic threat responses by firewalls or centralized management systems.
- Intrusion Prevention Systems (IPS)
  - At network level blocks traffic;
  - At host level kills processes, quarantines a file, blocks device access, etc...

Signature vs. Anomaly Based

- Intrusions are detected based on two different approaches:
  - Signature based:
    - Monitored data compared to preconfigured and predetermined attack patterns known as signatures;
    - Attacks have distinct known signatures;
    - Signatures must be constantly updated to mitigate emerging threats.
    - Signatures may contain:
      - Individual packet header values or binary data patterns,
      - Sequence of packets with specific characteristics within the same flow, or
      - Set of data flows (data stream) with specific characteristics (of flows or transmitted packets/data).
  - Anomaly based:
    - Establishes a behavior baseline (profile) and detects deviations from that profile;
    - May rely only on high-level systems or network statistics, or include multiple data sources;
    - May be based on predefined rules or on AI models.

Host-Based vs. Network-Based

- Host-Based - Endpoint Detection and Response (EDR)
  - To protect specific servers or user devices the IDS/IPS is deployed at the host level.
  - Monitors traffic, processes, files' access, devices' access and data flows, memory allocations, physical device characteristics (temperature, power consumption, movement, etc...).
- Network-Based - Network Detection and Response (NDR)
  - To protect an organization (all devices and services) the IDS/IPS is deployed at the network level.
  - Monitors traffic at the packet and flow levels. May monitor network at the physical level (radio, electric and optical signals).
  - Deployed at multiple network points:
    - Internet and WAN accesses;
    - Inter-zone communication links;
    - Wireless.

Endpoint Detection and Response (EDR)

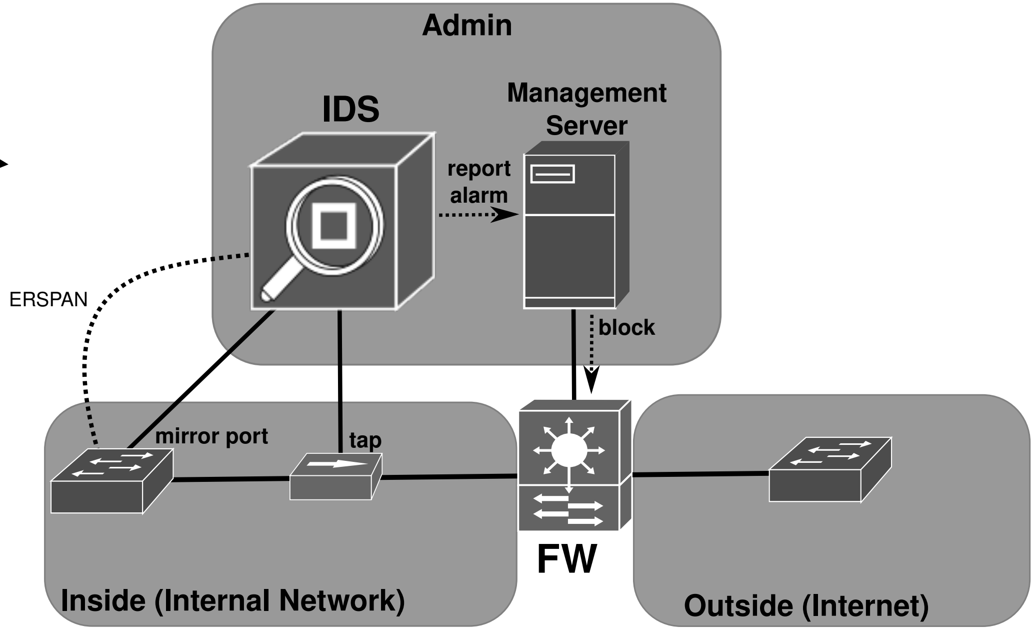
- Referred also to as endpoint detection and threat response (EDTR).
- Monitor, record and analyze the activities and events on devices.
- Provide continuous and comprehensive visibility of the devices processes and user activities.
- Enables a direct response to incidents in devices/servers.
- May be fully deployed only on the device, or with an agent on device and external data analysis/storage.

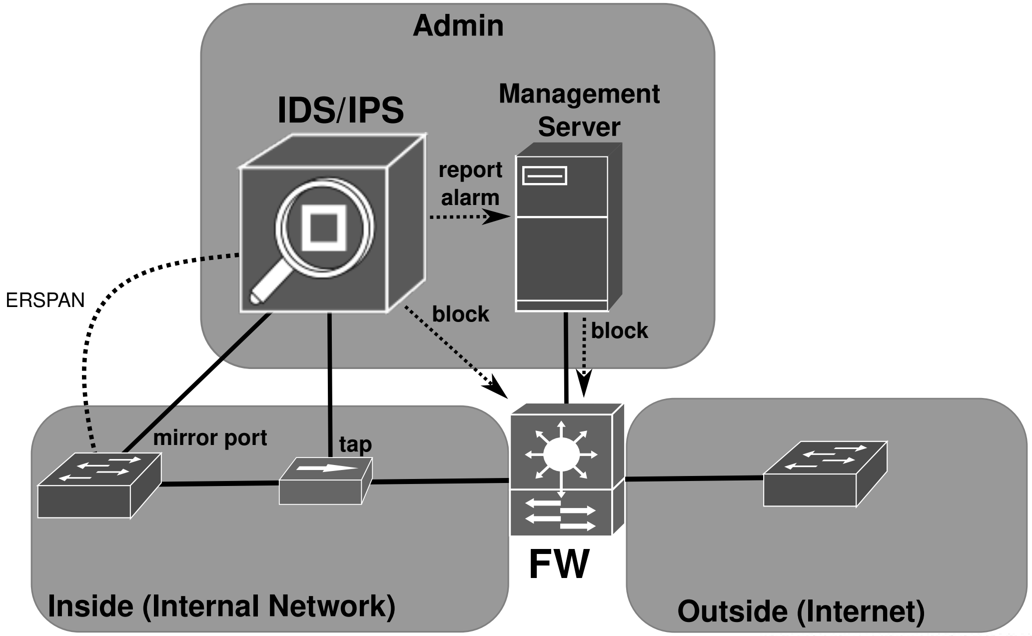
Network Detection and Response (NDR)

- Continuously monitors network communications.
- If an attacker has evaded log and host-based detection, then network analysis is the next and last resource to detected and mitigate attack.
- Very high volume of data to analyze and store.
- Requires dedicated network monitoring devices and methodologies.
    - Can be optimized by deploying pre-processing and processing data methodologies at each monitoring node.
    - Distributed computing can leverage data processing of high volume data.
- Current deployment based mainly on traditional IDS/IPS that use (localized) traffic flow data.
- Converging to a distributed system that gathers more detailed traffic data (packet data - requires data deciphering of own services data, flow dynamics over time, etc…)
    - Hard to deploy in terms of architecture, data volume and required processing power.
    - It is possible to offload workload to cloud services, but requires data pre-processing to reduce data transfers.

Network Deployment (1)

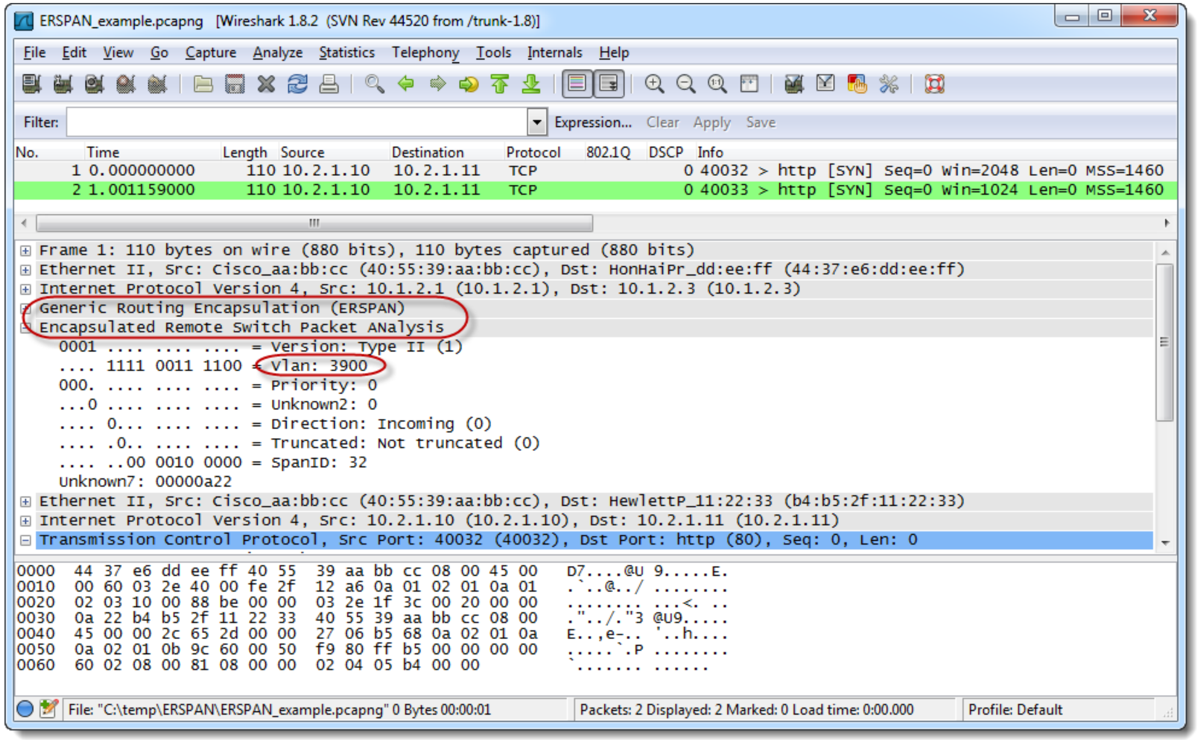
- IDS
  - Network tap.
  - Switch mirror port.
  - ERSPAN GRE tunnel from switch.
    - ERSPAN: Encapsulated Remote Switched Port ANalyzer
  - Reports to network management system.
- IPS
  - IDS with firewall integration.





ERSPAN

- Stands for “Encapsulated Remote Switched Port Analyzer”.
- Mirrors traffic from one or more switch ports.
- Sends the mirrored traffic to one or more destinations.
- The traffic is encapsulated in Generic Routing Encapsulation (GRE).



Network Deployment (2)

- IPS
  - Inline with firewall integration.
  - Inline with embedded firewall.